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ORIGINAL ARTICLE

Evaluation of the effect of empagliflozin on prevention of atrial fibrillation after heart valve surgery: a double-blind, randomized, placebo-controlled trial

Mahdi YazdFazeli[1] · Sahar Samimi[2] · Mohammad Abbasi Teshnizi[3] · Hamid Hoseinikhah[3] · Aliasghar Moeinipour[3] · Omid Javedanfar[3] · Vahid Ghavami[4] · Farshad Abedi[1] · Shahram Amini[5] · Benyamin Fazli[5] · Amir Hooshang Mohammadpour[1]

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Abstract

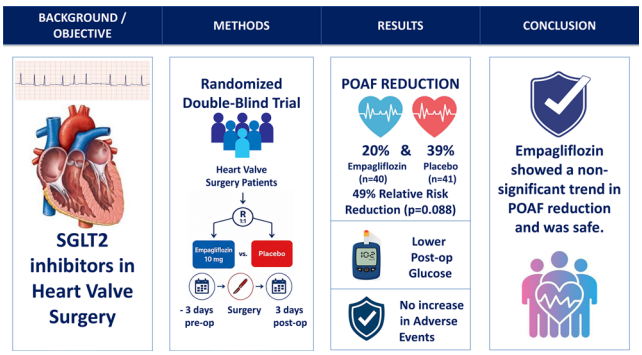
Background Postoperative atrial fibrillation (POAF) commonly complicates heart valve surgery. Empagliflozin, a sodium-glucose cotransporter 2 (SGLT2) inhibitor, may offer antiarrhythmic benefits. This trial evaluated its efficacy in preventing POAF.

Methods In a double-blind placebo-controlled trial, 110 patients undergoing elective heart valve surgery were randomized to oral empagliflozin 10 mg daily ($n=55$) or placebo ($n=55$) from 3 days pre-op to 3 days post-op. All patients received standard care. The primary analysis was performed in a modified intention-to-treat (mITT) population, defined as randomized patients with at least one assessable primary outcome measurement; 81 patients were included in the final mITT analysis. The primary outcome was POAF incidence within 72 h, assessed via continuous electrocardiography (ECG) monitoring. Secondary outcomes included other arrhythmias, C-reactive protein (CRP) levels, and glycemic control.

Results POAF occurred in 8 patients (20.0%) in the empagliflozin group and 16 patients (39.0%) in the placebo group, corresponding to a 49% relative risk reduction, although this difference did not reach statistical significance ($p=0.088$). Empagliflozin was associated with non-significant reductions in ventricular tachycardia (2.5% vs. 14.6%, $p=0.109$) and premature ventricular contractions (45.0% vs. 65.9%, $p=0.075$). Postoperative blood glucose levels were significantly lower in the empagliflozin group on days 1 and 2 ($p<0.05$). No significant differences were observed in postoperative CRP levels ($p=0.12$) or adverse event rates, including hypoglycemia and infection.

Conclusion Short-term perioperative empagliflozin did not significantly reduce the incidence of POAF after heart valve surgery; however, the results suggest a possible reduction in POAF and improved postoperative glycemic control, without an apparent increase in adverse events in this selected, closely monitored population. These findings require confirmation in larger randomized trials.

Graphical abstract



Keywords Heart valve surgery · Empagliflozin · Atrial fibrillation · Postoperative atrial fibrillation · POAF · SGLT2 inhibitor

Extended author information available on the last page of the article

Introduction

Postoperative arrhythmias, particularly postoperative atrial fibrillation (POAF), are frequent complications following heart valve surgery, contributing substantially to increased morbidity, mortality, prolonged hospitalization, and healthcare costs [1]. Among these, POAF has a reported incidence exceeding 30–50% following heart valve replacement surgery, which is significantly higher than the incidence observed after coronary artery bypass grafting (CABG) [2–4]. This arrhythmia typically manifests within the first five postoperative days, peaking between 24 and 72 h, and is associated with a fourfold increased risk of stroke and other serious complications [5–8].

The pathogenesis of POAF is multifactorial, involving an interplay of acute surgical stressors and chronic patient factors. Acute triggers specific to valve surgery include cardiopulmonary bypass (CPB)–induced systemic inflammation, atrial ischemia, oxidative stress, sympathetic activation, and electrolyte shifts [9–12]. Chronic substrates encompass structural and electrical remodeling of the atria, driven by conditions such as hypertension, diabetes, and heart failure, which are often present in this patient population. These mechanisms collectively illustrate that atrial fibrillation (AF) and POAF result from a complex interaction of structural, electrophysiological, autonomic, and inflammatory pathways. Perioperative factors such as fluid shifts, atrial stretch, hemodynamic instability, and sympathetic activation further amplify this vulnerability during cardiac surgery. Given the multifactorial nature and evolving understanding of POAF pathophysiology, ongoing investigation into novel preventive strategies remains essential [1, 13–16].

Sodium-glucose cotransporter 2 (SGLT2) inhibitors, including empagliflozin, are cardioprotective agents originally developed for type 2 diabetes [17]. Their benefits now extend to heart failure management irrespective of diabetic status, as reflected in current guidelines [18, 19]. A growing body of preclinical and clinical evidence suggests that empagliflozin may possess direct and indirect antiarrhythmic properties. Proposed mechanisms include modulation of cardiac ion channels (sodium and calcium), attenuation of inflammation and oxidative stress, improvement of endothelial and mitochondrial function, and reduction of atrial pressure and fibrotic remodeling [20–22].

Notably, a recent randomized trial found that perioperative empagliflozin significantly reduced ventricular arrhythmias and showed a non-significant trend towards lower POAF incidence in patients undergoing off-pump CABG [23]. However, the pathophysiology of POAF may differ fundamentally between CABG and valve surgery, with volume/pressure overload and progressive atrial fibrosis playing a more dominant role in the latter. Furthermore, the nearly universal use of CPB in valve surgery creates a pronounced inflammatory and pro-arrhythmic milieu. To date, no study has specifically investigated the role of SGLT2 inhibitors for POAF prevention in the high-risk context of heart valve surgery.

Therefore, we conducted this randomized, placebo-controlled trial to evaluate the efficacy and safety of short-term perioperative empagliflozin for the prevention of POAF and other arrhythmias in patients undergoing heart valve surgery.

Materials and methods

Ethical considerations and trial registration

This was a single-center, double-blind, randomized, placebo-controlled trial conducted at Imam Reza Hospital in Mashhad, Iran, between November 2022 and November 2024. The study protocol received approval from the Ethics Committee of Mashhad University of Medical Sciences (approval code IR.MUMS.REC.1401.129) and was prospectively registered with the Iranian Registry of Clinical Trials (registration identifier IRCT20120520009801N9). Written informed consent was obtained from all participants before enrollment, with assurances of data confidentiality and the right to withdraw at any time.

Study participants

Eligible patients were adults aged 18 to 70 years scheduled for elective heart valve replacement surgery. All patients included in the study were in sinus rhythm prior to surgery, as confirmed by preoperative electrocardiography (ECG). Key exclusion criteria were as follows: presence of a permanent pacemaker, severe heart failure (left ventricular ejection fraction < 30%), chronic kidney disease (estimated glomerular filtration rate < 20 mL/min/1.73 m^2), hepatic impairment (liver enzymes > 3 times the upper limit of normal), recent use (within 2 weeks) of any SGLT2 inhibitor, underweight status (body mass index ≤ 18.5 kg/m^2), history of previous cardiac surgery, pregnancy, or lactation, history of AF or atrial flutter, and concomitant surgical ablation. Patients for whom no assessable primary outcome data were available, or who developed predefined safety conditions preventing continuation of outcome assessment, were excluded from the modified intention-to-treat (mITT) analysis.

Randomization and blinding

Participants were randomly assigned in a 1:1 ratio to either the empagliflozin or placebo group. A computer-generated randomization sequence was created using variable block sizes (4 and 6) via the online platform www.sealedenvelope.

com. To ensure allocation concealment, an independent pharmacy technician, who had no role in patient enrollment or clinical assessment, prepared sequentially numbered, opaque containers according to this sequence. Each container held a 6-day supply of either empagliflozin 10-mg tablets (Gloripa®; Abidi Pharmaceutical Company, Tehran, Iran) or visually identical placebo tablets manufactured by the Pharmaceutics Laboratory of Mashhad University of Medical Sciences. Therefore, patients, treating clinicians (surgeons, cardiologists, and anesthesiologists), nursing staff, outcome assessors, and data collectors remained blinded to treatment assignment throughout the trial period. The blinding code was only revealed after the database lock and completion of the pre-specified statistical analysis.

Intervention

Patients allocated to the intervention group received oral empagliflozin 10 mg once daily. Those in the control group received a matching placebo on the same schedule. The study medication was initiated 3 days before the scheduled surgery and continued for the first three postoperative days (for a total of six doses). All participants received standard perioperative medical therapy and surgical care according to institutional guidelines, which could include beta-blockers, statins, angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers, and antiplatelet agents as indicated.

Study procedures and outcome assessment

Baseline demographic, clinical, and laboratory data were collected for all participants. Standard preoperative evaluations included ECG, transthoracic echocardiography, and selective coronary angiography. All surgeries were performed under general anesthesia using a standardized protocol involving midazolam, sufentanil, pancuronium, and propofol, with CPB.

Following surgery, patients were transferred to the cardiac intensive care unit (ICU) for continuous ECG monitoring for a minimum of 72 h. All arrhythmic events were reviewed and verified by a cardiologist who was blinded to treatment allocation. The primary outcome was the incidence of POAF, defined as any episode of AF lasting at least 5 min, with or without symptoms, occurring within the first 72 h post-surgery.

Secondary outcomes included the following: (1) incidence of other supraventricular arrhythmias (atrial flutter, atrial tachycardia); (2) incidence of ventricular arrhythmias (ventricular tachycardia (VT), frequent premature ventricular contractions (PVC)); (3) occurrence of new heart blocks; (4) changes in the inflammatory marker C-reactive protein (CRP), measured preoperatively and on postoperative day 3 using an enzyme-linked immunosorbent assay; (5) postoperative blood glucose levels; (6) durations of ICU and total hospital stay; and (7) safety outcomes, including hypoglycemia, urinary tract infection, and euglycemic diabetic ketoacidosis (eDKA). Patients underwent routine postoperative safety monitoring consisting of daily assessment of blood glucose levels, electrolytes, renal function, and complete blood counts, along with regular clinical evaluation and vital-sign monitoring. Safety assessment for suspected eDKA included arterial blood gas analysis when clinically indicated.

Sample size calculation

According to the previous studies, the prevalence of AF in patients undergoing heart valve surgery is estimated at 35% [3, 4]. Therefore, a sample size of 40 participants per group was calculated to detect a 25% reduction in the incidence of AF, using a two-sided test with 80% power and a 5% type I error. Considering an anticipated 35% dropout rate, the final sample size was set at 55 participants per group.

Statistical analysis

The primary analysis was performed in the mITT population, defined as all randomized participants with at least one assessable primary outcome measurement for POAF. Participants with no assessable primary outcome data were excluded from the mITT analysis.

Data analysis was conducted with SPSS software (version 24.0; IBM Corp.). Continuous variable distribution was evaluated via the Shapiro-Wilk normality test. For normally distributed parameters, results are reported as mean ± standard deviation, with group comparisons made by an independent sample *t*-test. Data not conforming to a normal distribution are expressed as median with interquartile range and analyzed using the Mann-Whitney *U* test. Within-group comparisons of non-parametric data were performed using the Wilcoxon signed-rank test. Categorical data are summarized as counts and percentages; differences between groups were assessed with the chi-square test or Fisher's exact test, depending on appropriateness. Statistical significance was defined by a two-sided *p*-value below 0.05.

Results

Study population and baseline characteristics

Between November 2022 and November 2024, 110 patients undergoing elective heart valve surgery were enrolled and randomly assigned to receive either empagliflozin ($n = 55$)

or placebo ($n=55$). Of these, 29 patients had no assessable primary outcome data because they were withdrawn before surgery or immediately postoperatively prior to any ECG monitoring for POAF and were therefore not included in the mITT analysis. In the empagliflozin group, 15 patients were excluded from the mITT population (4 due to increased serum creatinine and 11 due to incomplete essential study data), while in the placebo group, 14 patients were excluded (6 due to increased serum creatinine and 8 due to incomplete essential study data). Consequently, 40 patients in the empagliflozin group and 41 patients in the placebo group were included in the primary mITT analysis. The flow of participants is shown in Fig. 1.

The two groups were well-balanced at baseline. As shown in Table 1, no statistically significant differences were observed in demographic characteristics, medical history, or medication use (all $p>0.05$).

Similarly, Table 2 demonstrates no significant differences in echocardiographic parameters, laboratory values, or the type of valve surgery performed (all $p>0.05$).

POAF

During the 72-h postoperative monitoring period, POAF occurred in 8 patients (20.0%) in the empagliflozin group and 16 patients (39.0%) in the placebo group. This represents a 49% relative risk reduction associated with empagliflozin; however, the difference did not reach statistical significance ($p=0.088$; Table 3). No major protocol deviations were observed; therefore, the per-protocol population was identical to the mITT population and yielded identical results.

Other cardiac arrhythmias

Empagliflozin was associated with numerical reductions in the incidence of ventricular arrhythmias. VT occurred in 1 patient (2.5%) in the empagliflozin group versus 6 patients (14.6%) in the placebo group ($p=0.109$). Frequent PVC were observed in 18 patients (45.0%) and 27 patients (65.9%), respectively ($p=0.075$). No significant differences

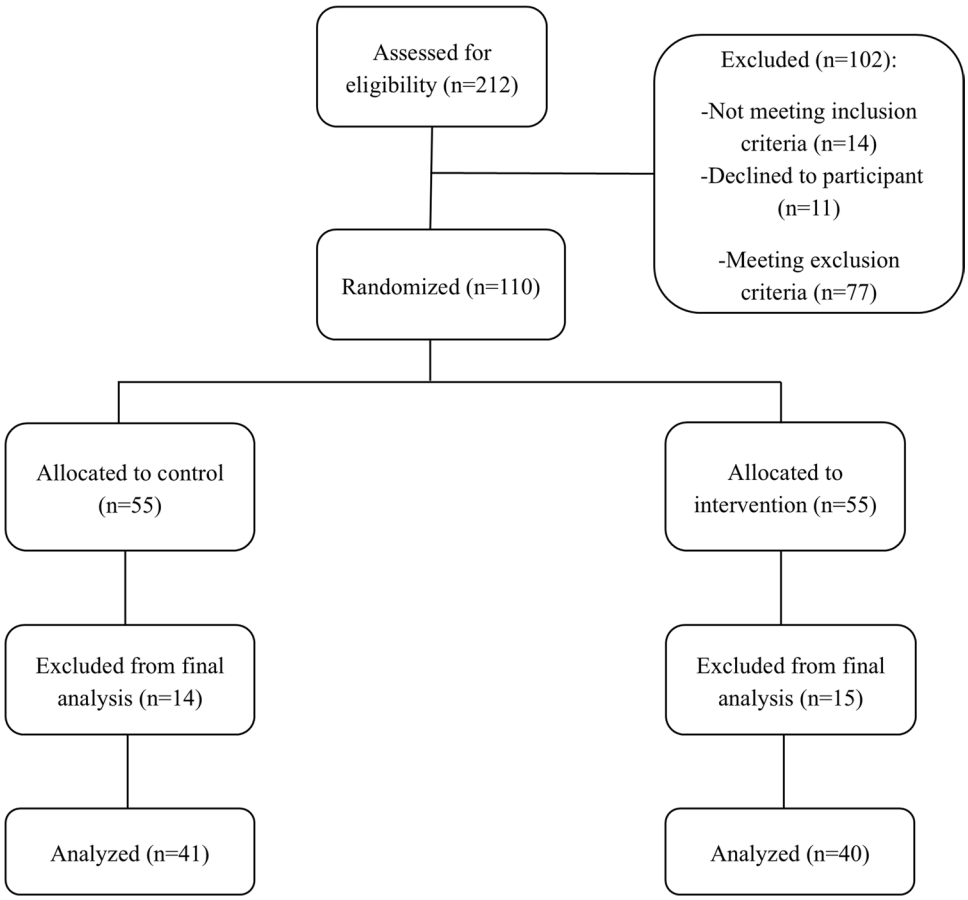


Fig. 1 Participant flow diagram

Table 1 Baseline demographic and clinical characteristics of the study participants

Variable	Empagliflozin ($n=40$)	Placebo ($n=41$)	p-value
Demographics			
Age (year)	52.10 (46–62)	53.73 (47–62)	0.58†
Male gender	22 (55%)	25 (61%)	0.59¶
BMI (kg/m^2)	25.26 ± 5.20	23.49 ± 3.69	0.08$
Substance history			
Smoking or tobacco	11 (27.5%)	17 (41.5%)	0.24¶
Alcohol	0 (0%)	1 (2.4%)	1.00¶
Opioid	8 (20%)	14 (34.1%)	0.12¶
Medical history			
IHD	10 (25%)	11 (26.8%)	1.00¶
MI	2 (5%)	2 (4.9%)	1.00¶
DM	7 (17.5%)	8 (19.5%)	1.00¶
DLP	10 (25%)	10 (24.4%)	1.00¶
HTN	18 (45%)	19 (46.3%)	1.00¶
Hypothyroidism	2 (5%)	2 (4.9%)	1.00¶
Medication history			
ACEI	9 (22.5%)	7 (17.1%)	0.58¶
ARB	6 (15%)	9 (22%)	0.56¶
β-blocker	22 (55%)	21 (51.2%)	0.82¶
Statin	14 (35%)	13 (31.7%)	0.81¶
CCB	4 (10%)	8 (19.5%)	0.34¶

Data are presented as *n* (%), median (IQR), or mean ± SD

†Mann-Whitney *U* test; ¶chi-square test; $independent sample *t*-test

ACEI, angiotensin-converting enzyme inhibitor; *ARB*, angiotensin receptor blocker; *BMI*, body mass index; *CCB*, calcium channel blocker; *DLP*, dyslipidemia; *DM*, diabetes mellitus; *HTN*, hypertension; *IHD*, ischemic heart disease; *MI*, myocardial infarction

were found between the groups in the occurrence of other supraventricular arrhythmias (atrial flutter, atrial tachycardia, premature atrial contractions), new-onset bundle branch blocks, or atrioventricular blocks (all $p > 0.05$; Table 4).

Inflammatory marker and glycemic control

Preoperative CRP levels were similar between the groups ($p = 0.61$). While CRP levels increased significantly by postoperative day 3 in both groups ($p < 0.001$ for both), the median CRP level was lower in the empagliflozin group (114 [86–163] mg/L) compared to the placebo group (155 [120–202] mg/L), though this difference was not statistically significant ($p = 0.12$) (Table 5).

In contrast, perioperative empagliflozin administration led to significantly better postoperative glycemic control. Blood glucose levels were significantly lower in the empagliflozin group on postoperative day 1 (median 162 vs. 191 mg/dL, $p = 0.02$), day 2 (140 vs. 163 mg/dL, $p = 0.01$), while on postoperative day 3 they showed a borderline decrease (118 vs. 130 mg/dL, $p = 0.05$), despite similar preoperative fasting blood sugar levels ($p = 0.94$) (Table 6).

Hemodynamics, clinical course, and other laboratory parameters

No significant differences were observed between the groups in average daily systolic/diastolic blood pressure, heart rate, or respiratory rate during the first three postoperative days. The net fluid balance was more negative in the empagliflozin group on postoperative day 2 ($p = 0.05$) but not on days 1 or 3. Duration of mechanical ventilation, ICU length of stay, and total hospital length of stay were comparable between the groups (all $p > 0.05$) (Table 7).

Among other laboratory parameters, platelet counts were consistently and significantly higher in the empagliflozin group on postoperative days 1, 2, and 3 (all $p < 0.05$). No significant intergroup differences were observed in white blood cell count, serum urea, or creatinine levels during this period (Table 8).

Safety outcomes

The perioperative administration of empagliflozin was well-tolerated with a safety profile comparable to placebo. No cases of eDKA or urinary tract infection were reported

Table 2 Echocardiographic parameters, laboratory data, and surgical characteristics

Variable	Empagliflozin ($n=40$)	Placebo ($n=41$)	p-value
Echocardiographic parameters			
LVEF	55 (50–55)	50 (41–55)	0.28†
LVSD	13 (32.5%)	14 (34.1%)	0.61¶
LVDD	10 (25%)	6 (14.6%)	0.32¶
RVSD	4 (10%)	12 (29.3%)	0.82¶
RVDD	1 (2.5%)	0 (0%)	0.49¶
LA enlargement	28 (70%)	28 (70.7%)	0.72¶
RA enlargement	5 (12.5%)	6 (14.6%)	0.36¶
MR	19 (57.5%)	21 (51.2%)	0.62¶
MS	12 (30%)	11 (27.5%)	0.39¶
AR	15 (37.5%)	21 (51.2%)	0.26¶
AS	12 (30%)	15 (37.5%)	0.55¶
TR	11 (27.5%)	12 (29.3%)	0.52¶
PR	1 (2.5%)	3 (9.7%)	0.15¶
Laboratory data			
WBC (10^3/μL)	7.7 (6.52–9.40)	7.6 (6.60–10.10)	0.88†
Hgb (g/dL)	12.98 ± 2.07	13.34 ± 1.67	0.39$
Platelet (10^3/μL)	209 (176–243)	179 (164–230)	0.11†
Urea (mg/dL)	30 (23–36)	25 (23–35)	0.28†
Sr Cr (mg/dL)	1 (0.9–1.2)	1.1 (0.9–1.2)	0.70†
FBS (mg/dL)	113 (91–157)	110 (93–145)	0.94†
Type of surgery			
MVR	24 (60%)	22 (53%)	0.56¶
AVR	21 (52%)	23 (56%)	0.82¶
TVR	4 (10%)	8 (19.5%)	0.34¶
PVR	1 (2.5%)	3 (7.5%)	0.61¶
Single valve	32 (80%)	30 (73.2%)	0.81¶
Double valves	6 (15%)	8 (19.5%)	0.78¶
Triple valves	2 (5%)	3 (7.3%)	0.73¶
Plus CABG	9 (22.5%)	8 (19.5%)	0.79¶
Valve type			
Mechanical	36 (90%)	34 (83%)	0.52¶
Bioprosthetic	4 (10%)	7 (17%)	0.52¶

Data are presented as *n* (%), median (IQR), or mean ± SD

†Mann-Whitney *U* test; ¶chi-square test; $independent sample *t*-test

AR, aortic regurgitation; *AS*, aortic stenosis; *AVR*, aortic valve replacement; *CABG*, coronary artery bypass grafting; *FBS*, fasting blood sugar; *Hgb*, hemoglobin; *LA*, left atrial; *LVDD*, left ventricular diastolic dysfunction; *LVEF*, left ventricular ejection fraction; *LVSD*, left ventricular systolic dysfunction; *MR*, mitral regurgitation; *MS*, mitral stenosis; *MVR*, mitral valve replacement; *PR*, pulmonary regurgitation; *PVR*, pulmonary valve replacement; *RA*, right atrial; *RVDD*, right ventricular diastolic dysfunction; *RVSD*, right ventricular systolic dysfunction; *Sr Cr*, serum creatinine; *TR*, tricuspid regurgitation; *TVR*, tricuspid valve replacement; *WBC*, white blood cell count

Table 3 Proportion of postoperative atrial fibrillation (POAF) between the treatment and placebo groups

	Empagliflozin ($n=40$)	Placebo ($n=41$)	p-value
Total POAF ($n=81$)	8 (20%)	16 (39%)	0.088¶

POAF, postoperative atrial fibrillation

Data are presented as *n* (%)

¶Data were analyzed by Fisher's exact test

in either group. Two episodes of hypoglycemia (5%) were recorded in each group, with no significant difference between them ($p=1.00$).

Discussion

In this first randomized controlled trial (RCT) investigating the role of an SGLT2 inhibitor in patients undergoing heart valve surgery, we found that short-term perioperative empagliflozin appeared safe in this selected, closely monitored population, but did not significantly reduce the incidence of POAF or other arrhythmias. However, several notable findings emerged: a clinically meaningful 49% relative reduction in POAF incidence (20.0% vs. 39.0%, $p=0.088$), numerical reductions in ventricular arrhythmias, and significantly better postoperative glycemic control with empagliflozin (Fig. 2).

The observed strong trend towards a lower POAF rate aligns with the proposed cardioprotective mechanisms of SGLT2 inhibitors, which include attenuation of inflammation, oxidative stress, and electrophysiological remodeling [24–26]. Our finding of lower, albeit non-significant, CRP levels in the empagliflozin group supports a potential anti-inflammatory effect. The pronounced numerical reduction in VT (2.5% vs. 14.6%) is particularly intriguing and echoes the significant reduction in ventricular arrhythmias demonstrated in the Dapagliflozin And Prevention of Adverse-outcomes in Heart Failure (DAPA-HF) trial with dapagliflozin [21] and in the prior study of empagliflozin in off-pump CABG patients [23]. Our results add a nuanced perspective to the evolving literature on SGLT2 inhibitors and arrhythmias. While meta-analyses in chronic heart failure and diabetes populations show benefits for atrial and ventricular arrhythmias [27, 28], a recent large meta-analysis found no significant prevention of AF [29]. Our study, situated in the acute, high-inflammatory setting of on-pump valve surgery, may reflect this complexity. The pathophysiology of POAF after valve surgery is heavily influenced by atrial pressure/volume overload

Table 4 Proportion of postoperative cardiac arrhythmias between the treatment and placebo groups

	Empagliflozin ($n=40$)	Placebo ($n=41$)	p-value
Ventricular tachycardia	1 (2.5%)*	6 (14.6%)*	0.109¶
Premature ventricular contraction	18 (45%)*	27 (65.9%)*	0.075¶
Atrial flutter	1 (2.5%)*	2 (4.9%)*	1.00¶
Atrial tachycardia	6 (15%)*	2 (4.9%)*	0.155¶
Premature atrial contraction	6 (15%)*	6 (14.6%)*	1.00¶
Right bundle branch block	4 (10%)*	9 (22%)*	0.226¶
Left bundle branch block	4 (10%)*	3 (7.3%)*	0.712¶
Atrioventricular block degree 1	2 (5%)*	3 (7.3%)*	1.00¶

*Data are presented as *n* (%)

¶Data were analyzed by Fisher's exact test

Table 5 Preoperative and postoperative C-reactive protein (CRP) levels

CRP	Empagliflozin	Placebo	p-value
Before intervention	2 (0.7–5.8)**	3.6 (0.3–8.9)**	0.61†
After intervention	114 (86–163)**	155 (120–202)**	0.12†
Within-group	$p<0.001$#	$p<0.001$#	

CRP, C-reactive protein

**Data are median (interquartile range)

***Data are mean ± SD

†Data were analyzed by the Mann-Whitney *U* test

#Data were analyzed by Wilcoxon signed-rank test

Table 6 Preoperative and postoperative blood glucose levels

Blood sugar (mg/dL)	Empagliflozin	Placebo	p-value
FBS pre-op	113 (91–157)*	110 (93–145)*	0.94†
BS-day1	162 (132–177)*	191 (147–242)*	0.02†
BS-day2	140 (122–170)*	163 (141–187)*	0.01†
BS-day3	118 (107–136)*	130 (118–160)	0.05†
Within-group	$p<0.01$#	$p<0.01$#	

BS, blood sugar; *FBS*, fasting blood sugar

*Data are median (interquartile range)

†Data were analyzed by the Mann-Whitney *U* test

#Data were analyzed by the Wilcoxon signed-rank test

and fibrosis—processes that may require longer-term treatment to modify. Therefore, a brief 6-day perioperative course, while practical, may be insufficient to significantly alter the underlying structural substrate for AF, in contrast to its more acute effects on ventricular irritability or glycemic control.

A key and potentially practice-informing finding of our study is the favorable safety profile and superior glycemic control achieved with perioperative empagliflozin. No cases of eDKA, urinary tract infection, or excess hypoglycemia occurred. This is especially significant as current guidelines recommend withholding SGLT2 inhibitors 3–7 days before surgery due to eDKA concerns [30]. Our protocol, involving careful patient selection and monitoring, demonstrates that short-term continuation may be feasible. Moreover, empagliflozin effectively mitigated the typical postoperative hyperglycemic stress response, achieving significantly lower blood glucose levels than placebo. This ancillary benefit could simplify postoperative management and potentially improve other hyperglycemia-linked outcomes.

One notable finding was the higher baseline platelet count observed in the empagliflozin group compared with the placebo group (209 vs. 179×10^3/μL). Although this difference was not statistically significant ($p=0.11$), such baseline variability may partly explain the persistently higher platelet values observed during follow-up. Therefore, these findings should be interpreted with caution, as they may reflect baseline imbalance rather than a direct effect of empagliflozin.

Table 7 Postoperative hemodynamic parameters, fluid balance, and clinical outcomes between the treatment and placebo groups

	Empagliflozin	Placebo	*p*-value
Day1_SBP (mmHg)	120 ± 13***	125 ± 13***	0.12$
Day2_SBP	126 ± 11***	129 ± 13***	0.18$
Day3_SBP	125 ± 14***	128 ± 14***	0.26$
Day1_DBP (mmHg)	69 ± 10***	69 ± 9***	0.68$
Day2_DBP	74 ± 10***	79 ± 9***	0.47$
Day3_DBP	72 ± 11***	75 ± 10***	0.22$
Day1_HR (/min)	82 ± 13***	78 ± 9***	0.13$
Day2_HR	82 ± 12***	79 ± 11***	0.22$
Day3_HR	80 ± 10***	79 ± 11***	0.76$
Day1_RR (/min)	16 (14–18)**	17 (14–19)**	0.22†
Day2_RR	18 (16–21)**	18 (17–22)**	0.27†
Day3_RR	17 (16–21)**	19 (17–21)**	0.11†
Day1_balance (mL)	149 ± 545***	72 ± 513***	0.51$
Day2_balance	−225[(−537) to (−77)]**	−50[(−525) to (300)]**	0.05†
Day3_balance	−152[(−175) to (0)]**	−106[(−200) to (0)]**	0.97†
ICU stay (day)	2 (2–2) **	2 (2–3)**	0.17†
Hospital stays (day)	7 (6–8)**	8 (7–10)**	0.14†
Endotracheal intubation duration (h)	8 (6–11)**	8 (6–10)**	0.80†

SBP, systolic blood pressure; *DBP*, diastolic blood pressure; *HR*, heart rate; *RR*, respiratory rate; *ICU*, intensive care unit

*Data are presented as *n* (%)

**Data are median (interquartile range)

***Data are mean ± SD

$Data were analyzed by independent sample *t*-test

†Data were analyzed by the Mann-Whitney *U* test

¶Data were analyzed by the chi-square test

Table 8 Serial postoperative laboratory parameters between the treatment and placebo groups

	Empagliflozin	Placebo	*p*-value
Day1_urea	33 (24–38)**	30 (25–37)**	0.27$
Day2_urea	39 (31–50)**	40 (31–50)**	0.90$
Day3_urea	41 (31–55)**	39 (29–54)**	0.44$
Day1_Sr Cr	1.00 (0.80–1.10)**	1.00 (0.90–1.20)**	0.68$
Day2_Sr Cr	1.10 (0.90–1.27)**	1.10 (0.90–1.25)**	0.96$
Day3_Sr Cr	1.00 (0.90–1.20)**	0.90 (0.90–1.15)**	0.57$
Day1_WBC	16.00 (14.00–20.00)**	15.00 (12.00–18.00)**	0.07$
Day2_WBC	13.50 (11.32–16.75)**	13.00 (11.0–16.00)**	0.34$
Day3_WBC	14.00 (11.77–17.00)**	13.00 (11.00–16.50)**	0.44$
Day1_PLT	185 (141–210)**	141 (117–181)**	0.007$
Day2_PLT	160 (132–187)**	132 (116–158)**	0.030$
Day3_PLT	159 ± 44***	138 ± 38***	0.022†

Sr Cr, serum creatinine; *PLT*, platelet; *WBC*, white blood cell

**Data are median (interquartile range)

***Data are mean ± SD

$Data were analyzed by the Mann-Whitney *U* test

†Data were analyzed by an independent sample *t*-test

Comparing these results with a previous RCT in CABG patients is instructive [23]. In both studies, empagliflozin showed a non-significant trend towards reducing POAF. However, the signal for ventricular arrhythmia suppression was stronger and reached statistical significance in the CABG cohort. This difference may be attributable to the distinct pathophysiology; ischemic myocardium in CABG patients might be more immediately responsive to the metabolic and ion channel modulating effects of empagliflozin, whereas the arrhythmic substrate in valve disease is more chronically established.

Limitations

This study has several limitations. First, because 29 randomized patients had no assessable primary outcome data and were excluded from the mITT analysis, the findings are vulnerable to attrition bias and should not be interpreted as equivalent to a full intention-to-treat analysis. Second, although the target sample size was pre-specified based on an anticipated 25% relative reduction in POAF, the observed reduction in our study was approximately 19% points. In addition, missing

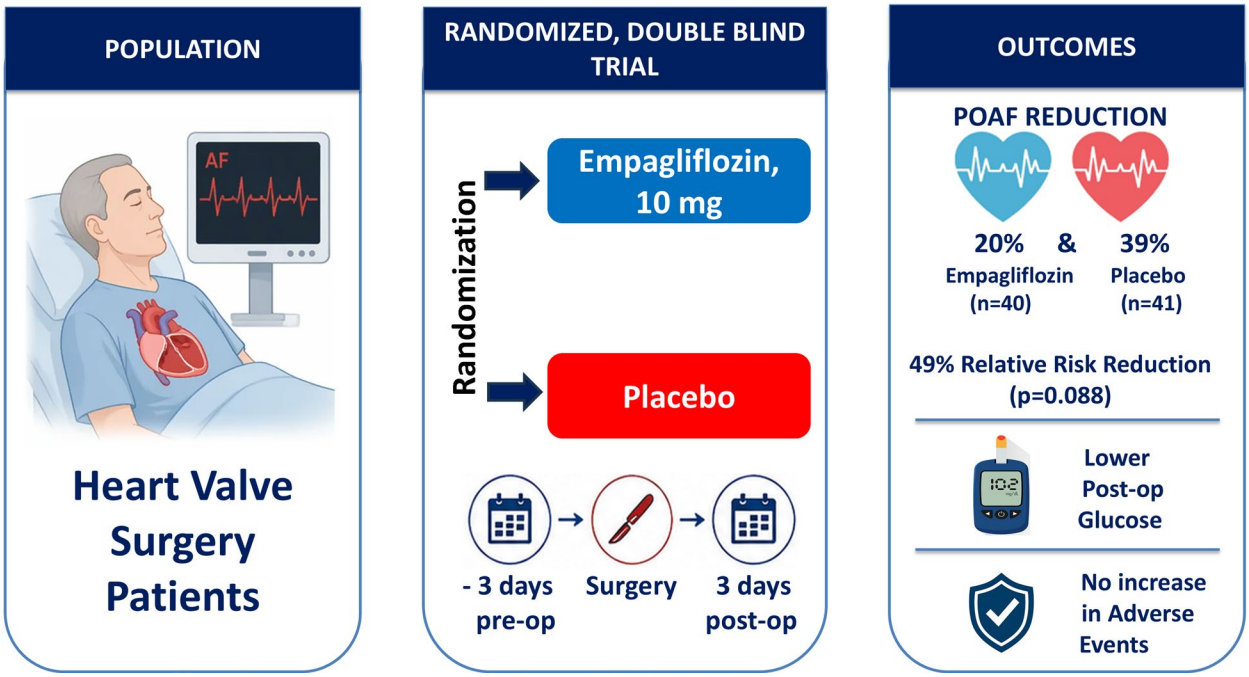


Fig. 2 Central image of the study results

primary outcome data led to a smaller mITT population than originally planned. Together, these factors may have reduced the statistical power of the study to detect significant differences in POAF. Third, although key demographic and standard echocardiographic variables were collected, more granular characterization of surgical details and the etiologic classification of valvular pathology was not available. Fourth, systematic assessment of safety outcomes such as hemodynamic instability, fluid balance, infection, and metabolic complications was limited by the sample size and available data. Fifth, the study was not powered to perform subgroup analyses, which could help identify populations most likely to benefit. Finally, only a single perioperative dosing strategy and a short treatment duration were evaluated, which may be insufficient to influence the chronic atrial substrate underlying POAF.

Conclusion

In this mITT analysis, perioperative empagliflozin appeared safe in this selected, closely monitored cohort and showed non-significant trends towards lower POAF and ventricular arrhythmias. Although empagliflozin improved postoperative glycemic control, these findings should not be interpreted as sufficient to alter current precautionary discontinuation strategies for SGLT2 inhibitors. The strong trends observed, especially when considered alongside our previous positive results in CABG, justify and indeed necessitate a larger, multicenter, adequately powered randomized trial. Future studies should consider longer pre-treatment durations, incorporate mechanistic endpoints, and further define the optimal and safe perioperative management of SGLT2 inhibitors to harness their full cardioprotective potential.

Author contribution Conceptualization and study design: AHMP, BF. Methodology: AHMP, BF, VG. Data collection: MYF, SS, MAT, HH, AM, OJ, SA, BF. Formal statistical analysis: VG. Writing—original draft: MYF, FA. Writing—review and editing: all authors. Supervision: AHMP, BF. All authors read and approved the final manuscript.

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Data availability The datasets generated during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval The study protocol was approved by the Ethics Committee of Mashhad University of Medical Sciences (IR.MUMS.REC.1401.129).

Informed consent Written informed consent was obtained from all participants prior to enrollment.

Competing interests The authors declare no competing interests.

Human and animal rights statement This study was conducted in accordance with the ethical standards of the institutional research committee and the Declaration of Helsinki.

AI disclosure statement The authors used Grammarly only for language editing and grammar checking. No artificial intelligence tools were used for data analysis, data interpretation, figure generation, or generation of scientific content.

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Authors and Affiliations

Mahdi YazdFazeli[1] · Sahar Samimi[2] · Mohammad Abbasi Teshnizi[3] · Hamid Hoseinikhah[3] · Aliasghar Moeinipour[3] · Omid Javedanfar[3] · Vahid Ghavami[4] · Farshad Abedi[1] · Shahram Amini[5] · Benyamin Fazli[5] · Amir Hooshang Mohammadpour[1]

✉ Benyamin Fazli
Fazlib@mums.ac.ir

✉ Amir Hooshang Mohammadpour
Mohamadpoorah@mums.ac.ir

[1] Department of Clinical Pharmacy, School of Pharmacy, Mashhad University of Medical Sciences, Mashhad, Iran

[2] Department of Cardiovascular Diseases, Faculty of Medicine, Mashhad University of Medical Sciences, Mashhad, Iran

[3] Department of Cardiac Surgery, Mashhad University of Medical Sciences, Mashhad, Iran

[4] Department of Biostatistics, Social Determinants of Health Research Center, Mashhad University of Medical Sciences, Mashhad, Iran

[5] Department of Anesthesia, School of Medicine, Mashhad University of Medical Sciences, Mashhad, Iran